Adapting the DisCoCat-Model for Question Answering in the Chinese Language

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***Abstract*—We introduce quantum natural language processing for the Chinese language. Our approach focuses on Question-answering on a set of sentences, whether a particular sentence is truthful with respect to the whole corpus of sentences. We employ the Categorical Distributional Compositional (DisCoCat) model to translate sentences to valid quantum circuits. We achieve a fitting score of 97% on the test set. Our sentence set is also significantly larger than previous experiments. The results show general applicability of the framework to other languages. We also show that it can be used to introspect natural language models and provide new approaches to model explainability.**

I. Introduction

Natural Language Processing (NLP) is ubiquitous in modern life. A wide variety of services from translation services, spam detection, chatbots and many other applications make use of an automated understanding of language [1]. It is especially important as its interface to human interaction. Where human-machine interaction is not limited to a fixed set of actions through a set interface, some form of natural language processing needs to be employed to accurately capture the requests of the operator. Understanding large bodies of text through summarization and extracting information from it is also a prime task for NLP [2].

Historically, several approaches to automated natural language processing have been tried. All rely on separating the full meaning of a document or sentence into its constituent parts, words. Formal semantics attempts to structurize meaning through structure, for example by expressing structure through some lambda calculus [3]. However, structure does not induce meaning. Alternative approaches have focused on modeling meaning, either through co-occurrence of words for similarity [4] or contextual modeling. Here a model is found that maximizes an objective function on the probability of surrounding words given a context [5]. Predicting and approximating is very natural to neural networks, so these neural word embeddings make use of large neural networks to learn and represent contextual information given a particular word or sentence. This approach ignores the underlying structure of language in the form of grammar in favour of modeling a contextual probability [6]. For small text samples these different models perform about equal for different tasks, but at scale neural networks outperform the other methods [6].

The performance of neural network based NLP is heavily reliant on its structure and training data. The probability model can vary and is also relevant to the success of the model, from bag-of-word style models [7], attention mechanisms from transformer-style architectures [8] to complex combinations of constructions [9]. A common criticism of these models is their explainability, where the model is effectively a big black box of numbers opaque to inspection. Poking at the black box with explorations focused the word embeddings themselves has led to interesting insights to the biases of the model [10], [11]. Some of the structures shown in natural language processing models reflect historic and contemporary biases present in the data [10]. Due to their probabilistic models, treating the biases is difficult and prone to error. The connected lack of explainability of neural networks drives a continuous effort to develop explainable models.

In particular the formal grammar approach is used to develop explainable models, as it formalizes the connections between words and provides a firm footing for reasoning about structure. In a surprising connection, the mathematical model underlying the formal grammar is similar to approaches in quantum computing.

A quantum computer is a technical system that, rather than representing information through classical bits, works with qubits. A computation on a quantum computer can make use of a much larger state space. This difference in accessible computational space is one of the reasons

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why large computational speedups are expected from quantum computing [12]. Information transfer into the system happens through state preparation, the output via a measurement process that collapses the wave function to a bit string again [12].

According to Coecke et al., "the fact that quantum theory and natural language [...] share the use of vector spaces for describing states – albeit for very different reasons – makes those two theories (essentially) coincide. Therefore we say that QNLP is 'quantum-native'" [13]. Not only do current QNLP approaches cover word meanings as well as grammar rules and are therefore more intuitive, Zeng et al. [14] have furthermore proven that a quadratic speed up over classical hardware is theoretically possible for certain QNLP applications.

Use cases that have been covered since this initial experiment are, among others, Question Answering, Sentence Similarity and Text Classification (e.g. [15], [16]). The majority of these implementations has been specifically tailored for the English language and achieved highly promising results. Only few experiments were thus far adapted for non-English languages (Italian [17], Hungarian [18], Sanskrit [19]) and none for Sino-Tibetan languages, such as Chinese.

The objective of this paper is to assess the applicability of the pregroup-based QNLP approach to Chinese, which has a different grammar structure compared to English. We achieve this through a question-answering scheme similar to [16], adapted to the Chinese language[1] and run on a quantum simulator: Short declarative sentences are mapped onto quantum circuits, their syntax being represented by certain gates and their word meanings by gate or state parameterization. The evaluation of the individual circuit returns the truth value for the corresponding sentence which can be viewed as the answer to a yes-or-no question. By adapting this experiment for the Chinese language and its distinct pregroup grammar, the practicability of the DisCoCat approach for languages with a different typology than English can be assessed.

II. Background

We introduce the necessary mathematical background for translating sentences into quantum circuits. The basic formalism is a pregroup algebra. The translation to quantum circuits hinges on the distributional combinatorial categories framework (DisCoCat). We also show its connection with question answering tasks. Finally we provide some insights into the differences between English and Chinese.

A. Pregroups

In this work we translate a sentence into a pregroup formalism. A pregroup is a partially ordered algebra $(A, 1, \cdot, {}^{-l}, {}^{-r}, \leq)$ such that $(A, 1, \cdot)$ is a monoid. This formal definition of a monoid describes a set of elements A with an identity 1 and an operation $\cdot$ between two elements of A. Different to a full group, a monoid does not possess an inverse [20], [21]. The pregroup expands the monoid by introducing a left inverse $-^l$ and right inverse $-^r$ that can selectively cancel operations but fall short of a true inverse necessary for a full group structure. Grammatical structures can then be translated to a pregroup elements. Consider the sentence "Alice loves Bob". From a grammatical perspective, there are two nouns, "Alice" and "Bob", connected by a verb. This relationship can be expressed in a pregroup formalism by assigning types to each word. A noun can be assigned the pregroup type n and verbs that link between a subject and object such as "to love" are identified as $n^r s n^l$. s here means a sentence token. Hence the grammar of the sentence can be expressed as $n \cdot n^r s n^l \cdot n$. A sentence is grammatically valid in the pregroup sense if the pregroup elements corresponding to the words cancel out to a sentence element. Different grammatical functions need to be assigned different pregroup types. The pregroup structures can be nested to reflect long-distance relationship and thereby express natural language grammar [22]. Having a remaining s element instead of contracting to an identity is useful for the following quantum computing construction.

B. DisCoCat model

The concept of translating between grammar structures and pregroups has also been translated into the language of category theory. It provides a formal basis to speak about equivalences between grammar, pregroups and, finally, quantum circuits. We forego an in-depth exploration of the topic and refer the interested reader to [20]. For the purposes of this work, the categorical framework allows for a formal homomorphism between the different structures grammar, pregroups and quantum circuits. On top of that lies the distributional compositional (DisCo of DisCoCat) framework, wherein the distributional inference of word meanings is introduced. Similar to the natural language frameworks such as bag-of-words mentioned above, the meaning of a individual word can be modeled in the context of its surroundings. This can be a word-vector embedding where meaning is indicated through relative distances to neighbouring vectors or some probabilistic framework. Hence the structure and grammatical meaning is represented by its corresponding pregroup element, but its meaning is affixed to the pregroup element by some form of parameters. Finally, compositional means that the structures can be composited into larger sentence-equivalents and this is true for their equivalent quantum circuits as well. Individual pregroup elements and therefore the grammatical structure can, in this framework, be principally represented by quantum circuits and these circuits can be composited in a meaningful way [23].

[1] In this paper, Chinese refers to Mandarin Chinese. The characters used are simplified Chinese characters and the Pinyin transcript is depicted without tones.

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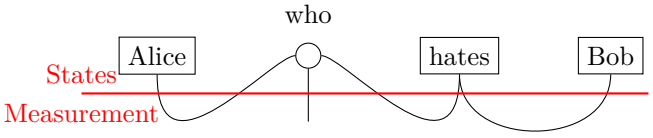


Fig. 1: The DisCoCat approach encorporates both semantics and syntax, being represented as different states and measurements on the quantum hardware.

These different layers can also be expressed in a form of graphical calculus where the pregroup structure is laid out in a linear fashion representing the sentence and lines linking the different contractions are drawn below it. On a quantum computer, the diagram can be formally interpreted as a series of states on qubits with the connecting links formed by quantum circuits. This is also formally similar to ZX calculus [24], which is used to reason about quantum circuits on the basis of graphical interaction rules. The different separate circuits are joined via Bell measurements [25]. A visual representation of this reasoning is shown in figure 1, with time flowing downward; First the quantum circuits representing the words are constructed, followed by a Bell measurement on the wires corresponding to the cancellations of the pregroup. On wire, carrying the information corresponding to the s type pregroup is then used to measure truthfulness of the sentence. Figure 1 shows this on an abstract level. The quantum circuits corresponding to each pregroup create a state, an Bell measurements on connected pregroups link the elements together. The last wire for s carries the truth value of the sentence.

In DisCoCat-based QNLP use cases, words can be implemented as vectors according to their distribution in a text corpus, for instance by counting their occurrence in relation to a set of preselected basis words. The syntax is represented using pregroup rules that are modeled according to the grammar rules of the language and mapped onto a parameterized quantum circuit. The semantics, on the other hand, are encoded within the parameters of the circuit. The parameters can be either defined manually or learned.

When executing a DisCoCat task on a quantum computer, the semantics of an individual word are described as quantum states, while the grammatical interdependence of the words corresponds to Bell measurements.

Both the syntax and the semantics of a sentence are taken into account for the implementation: The pregroup-based syntax of a sentence is mapped onto a parameterized quantum circuit, while the semantics are encoded within the parameters of the circuit. The parameters can be either defined manually or learned [26].

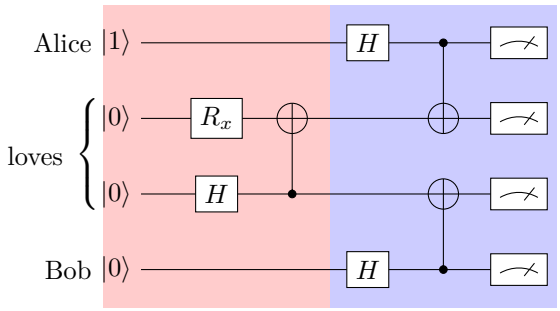


Fig. 2: Quantum circuit encoding the sentence "Alice loves Bob". The first part of the circuit marked in red prepares the states according to the pregroup. The second half marked in blue prepares a Bell measurement on the qubits.

C. Question answering

The general set up of the original Question Answering use case can be described as a special type of binary classification task [26]. A quantum gate computer or simulator is trained on a set of labeled sentences, where each label indicates the truth value of its corresponding sentence. The goal is, to be able to present an entirely new sentence (made of the previously learned vocabulary) to the quantum device and have it return the correct truth label for the sentence. We choose a set of words, listed with their pregroup identification in table I, and construct a corpus of sentences out of it. This is the basis for our training data. For many languages, the pregroup types for a grammatical element can be taken from literature. Alternatively one can construct the pregroup structure themselves. We have started from the English pregroups and introduced our own for grammar structures that have no corresponding structure in English. DisCoCat diagrams can now be generated and automatically parsed by using the Discopy [27] tool. Each pregroup element is translated to a parametrized quantum circuit. The obtained complete circuit corresponding to a sentence is then used to train the parameters for the word meanings. After the training is completed successfully, the quantum device can provide an answer to any question by evaluating the truth value or meaning of the given sentence. This truth value is measured on the qubit carrying the s pregroup type, which can be used as a truth value. In other words, the quantum computer is trained to interpret the meaning of a limited set of words and can – provided with the grammar rules of the language – respond to any yes-or-no question that can be composed of the chosen vocabulary.

III. Related Work

The theoretic foundation for most of current QNLP research was laid by Bob Coecke, Mehrnoosh Sadrzadeh and Stephen Clark in [28]. In this joint paper, their previously distinct areas of expertise were combined in a new approach: Coecke had been studying categorical quantum mechanics [29], Sadrazadeh was a renowned

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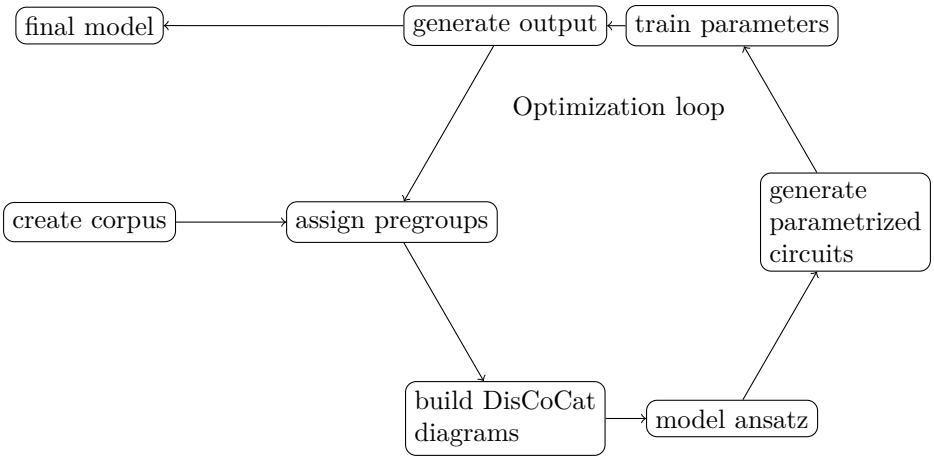


Fig. 3: Workflow for question-answering. Based on a corpus, there is an optimization loop between finding a suitable pregroup representation, building the circuit and training its parameters. The final model with its trained parameters can then be used for other tasks.

grammar algebra expert, largely basing her methods of parsing grammatical sentences on Lambek's mathematical formalisms for processing natural languages and syntactical structures [30], and Clark dealt mostly with vector-space NLP which he utilized as a way to encode the meaning of words [31].

By merging Clark's distributional theory of meaning and Sadrazadeh's Lambek-inspired compositional theory for grammatical types, they proposed a mathematical framework for encoding natural languages which was later further developed into the so-called DisCoCat model. This model views a sentence as a "network in which words interact in a particular fashion" instead of a sole "bag of words" [32]. This means that both the structure and the semantics of a sentence are said to contribute to its overall meaning. The DisCoCat model takes both of them into account and offers a way to represent and discern the meaning of a sentence by using tensor product composition on a quantum computer. This approach is the basis for the majority of QNLP research, as well as for this paper.

The viability of the DisCoCat concept was proven by Zeng et al. [14] who showed that with the framework a quadratic computational speedup over classical methods is possible. However, they assume the existence of Quantum Random Access Memory (QRAM) for their theoretic argumentation. Hereupon, Coecke et al. proposed a QNLP alternative for current quantum hardware without the need for QRAM, considering variational circuits instead [13]. Their first experiment (titled *Alice-loves-Bob*), a Question Answering use case based on the DisCoCat model that incorporates the python library `DisCoPy` (Distributional Compositional Python), was run on a classical computer and demonstrated at the QNLP2019 conference [33]. It encoded the meaning of a small sample of words as parameters of a quantum circuit and calculated the truth value for one single question. Later, this experiment was also successfully reproduced on an IBM quantum computer [34] [16] which not only proved the transferability of the DisCoCat approach onto actual quantum hardware but also laid the cornerstone for future QNLP research.

Numerous papers have since been published on the subject of QNLP. A first comprehensive literature survey and categorization of QNLP algorithms was presented by Wu et al. [35]. According to them, significant progress has been made regarding the two QNLP tasks Question Answering and Text Classification: In addition to the above-mentioned experiments by Coecke et al., Lorenz et al. [36] also made a pioneering contribution in the area of Text Classification. They managed to map a dataset of $\geq$100 sentences onto quantum circuits using the DisCoCat model and successfully executed a simple sentence classification task on the quantum computer `ibm_bogota`, attaining a test error of only 16.7%.

While there have been some promising classical or hybrid concepts for QNLP algorithms, [37]–[39], current research mainly focuses on applying Coecke's DisCoCat approach to diverse and more complex NLP topics. Recently, particular progress has been made for Sentence Similarity tasks. In [15], Abbaszade et al. propose an algorithm for sentence similarity use cases with positive, transitive sentences in the English and Persian language. Although they did not test this algorithm on quantum hardware, they prove the general transferability of the DisCoCat model to another language and draw a positive conclusion regarding future machine translation on quantum computers. However, they only use two sentences in their evaluation, greatly limiting the extrapolation possible from their results.

DisCoCat was also used in [40] as the basis for Sentence Similarity calculations for the Spanish and English language. The authors create equivalent English and Spanish language sentences based on 45 sentences. In

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their approach they model equivalent sentences and then compare the parameters as vectors to achieve a similarity score. Each corresponding word is mapped to the same dimension. They report an average cosine distance of .95 as their similarity score. Their approach differs from ours in that their training focuses on training parameters in comparison, rather than towards a single true/false label as used in this work.

Le Du et al. [26], on the other hand, explored the potential of the DisCoCat model by analyzing a variety of practical and theoretical QNLP use cases, for instance Binary Sentence Classification and Hyponymy (=hierarchical word relation). They managed to expand previous Question Answering implementations by adding three logical connectors, reaching an overall fitting score of 62.62% when running it on a quantum simulator. Furthermore, they drew a generally positive conclusion regarding the future potential of the DisCoCat model for other QNLP tasks, especially pointing out the progress made by Bankova et al. who adapted the categorical compositional distributional framework to account for hyponymial relationships between words [41].

Until now, QNLP – despite it being a comparatively new field of research – has seen a lot of progress, especially in recent years. To a large extent, this progress has been limited to Sentence Similarity and Questions Answering tasks. First practical DisCoCat-based applications on quantum hardware have shown promising results. Most of the research was done for the English language, yet in the field of Sentence Similarity, proposals for other languages (e.g. Spanish, Persian) were made as well. Thus far, no Question Answering experiments have been implemented for the Chinese Language.

IV. Methods

A. Differences between English and Chinese

A full-scale difference analysis of linguistical differences between the two languages is out of scope here. Instead, we focus on the grammatical differences as they are relevant to the DisCoCat model and the implementation here. Both languages follow a subject-verb-object order, with a verb connecting between the subject and the object. Chinese features verbless adjective descriptions, such as "鲍勃很高", translating as "Bob [is] very tall". Here there is no explicit verb linking subject and adjective description. Objectless verbs work similarly to English, with a subject-verb sentence such as "鲍勃睡觉", "Bob sleep[s]" working the same way as its English equivalent. [42], [43]

For the present paper, an important difference is the structure of connecting words such as "who", "which", "of",.... For the English language, these connecting words appear in a specific context, depending on the type and meaning of subject and referenced object. In Chinese, these words are all handled by the particle "的" whose translated meaning changes depending on the surrounding context in which it is employed [43], [44].

Chinese is different to the English language in its characters. Each lexical word is composed of one or more characters. This can be seen at a glance from table I, where different amounts of Chinese characters correspond to one meaning. Characters as a whole can be recombined to create new words and often carry an general meaning that is made concrete when put in conjunction with other characters. Different to English, words made up of characters also do not have spaces between them, so discerning word borders is not trivial. We forego this complication to separate the words and embed their meaning directly.

On the other hand, the character system also means that verb inflexion, where the verb changes its spelling depending on the expressed person, time or mood. In English, this is reduced to the -s ending for 3rd person singular and changes to the verb for past tense or passive voice. For many pregroup systems, the change with person is quietly ignored as it does not change the underlying grammatical function. In Chinese, no character changes and tense and mood are expressed by adding a word ("过") for past tense and changing the word order ("被") for passive voice.

The listed differences warrant the construction of our own pregroups.

B. Pregroup formalism

Creating a QNLP implementation for the Chinese language requires a pregroup formulation. There is no ready-made implementation for the Chinese language. We create our own to capture sentence structure of simple sentences, relative clauses, questions and comparative structures. We use small corpus of sentences of eight words together with five parameters, in line with other experiments [25]. Starting from the corpus, we build up a pregroup grammar to capture the different structure of the Chinese language. The words used in our experiments and their pregroup identification are listed in table I. The pregroup we define here is created with the experiment in mind and does not attempt to capture the whole grammatical structure of Chinese.

Chinese term	Translation	Pregroup
爱丽丝, 'ai li si'	Alice	n
鲍勃, 'bao bo'	Bob	n
爱, 'ai'	to love	$n^r s n^l$
恨, 'hen'	to hate	$n^r s n^l$
很小, 'hen xiao'	small	$n^r s$
很高, 'hen gao'	tall	$n^r s$
睡觉, 'shui jiao'	to sleep	$n^r s$
的, 'de'	*particle*	$s^r n^{rr} n n^l$

TABLE I: The corpus of Chinese words used in the experiments, with their English translations and corresponding pregroup identification.

For the purpose of the experiment, we focus on the larger differences to English. While the basic sentence structure of subject – verb – object is the same, Chinese allows for verbless subject – object sentences like

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"爱丽丝很小", "Alice [is] very small". We also implement non-object verbs ("鲍勃睡觉", "Bob sleeps"/"Bob is sleeping") and the particle "的", a connecting particle with no direct translation but roughly similar to English particle words such as "of". Each sentence is mapped to a truth value that indicates whether the particular sentence is truthful. This provides the training target for the quantum computation. Labeling of the sentences is consistent across the training set such that there is conflicting statement between two sentences.

Via the pregroup correspondence of a sentence, it is possible to construct a circuit ansatz. [23], [25], [45] Each pregroup element is identified with a quantum circuit element. The circuit then encodes the sentence structure. The meaning of the word is encoded in the parameters within the circuit block. The parameters are what is trained in the experiments using the example sentences and their truth values. The circuits are constructed on a Quantum cirucit simulator and trained using the already established SPSA optimizer [46]. For the experiment, five parameters are sufficient, which in turn is achievable on current simulators. The circuit elements corresponding to a pregroup element are not prescribed. The most common ones are combinations of CNOT gates together with single qubit rotations and combinations of Hadamard and CZ gates [47]. In principle, any ansatz can (be made to) work. We detail our circuit elements in figure 4. These are, with the exception of the "的" particle, direct translations of the corresponding ansatz for the English language as implemented in the DisCoPy package [27]. The connection between the different circuit elements is made using constructing Bell measurements on the connections stipulated by the pregroup grammar resolution rules. This is shown graphically for a complete circuit in figure 2. The individual circuit elements that we use are shown in figure 4.

Grammatical correctness can be verified on the grammar diagram and the circuit by checking that all but one qubit wire is part of a GHZ state [25]. Measuring this last qubit wire s can be interpreted as a truth value $\langle 0|s\rangle \in \{0, 1\}$. This measurement is also the training target for changing the parameters of the circuit.

V. Evaluation

A. Dataset properties

We construct our dataset out of the eight words listed in table I. We provide the full dataset with Chinese sentence, translation and truth value in the appendix, table III. Each sentence is connected to a truth value indicating whether the sentence is truthful or not. We take care not to include contradictions in our sentences. Not all words are present in all sentences; As a result, we have to take care to connect the training of parameters inside the circuit to the parameters corresponding to that word. Our dataset is almost exhaustive in the combinations between words. We also have a bias towards 'false' truth values due to the internal consistency requirement, with 15/50 true sentences. However, our results and training performance cannot be explained by always outputting 'false' and indeed we do not observe this.

B. Fitting score

We are evaluating the experiments on their achieved truthiness, i.e. whether a sentence is correctly labeled as true or false by the quantum circuit simulator. With 50 sentences and a 50% split between training and testing sentences, we achieve a score of 94% on the test set (97% on the training set) after 100 iterations of the SPSA optimization algorithm. This is in line with the results obtained for other languages, shown in table II.

	Chinese	English [25]	English [26]
# of sentences	50	16	20
Parameters	5	3	2
Hardware	simulator	quantum	simulator
Circuit depth	1	2	1
Fitting score	95%	n.s.	78%
Mean error	0.018	0.313	n.s.

TABLE II: Comparison between our experiment and corresponding experiments for the English language.

We also find that, compared to classical approaches such as bag-of-words [7], BERT [9], transformer architectures [8] and other forms of deep language models, we can provide a clear connection between the language meaning and the technical construction.

C. Explainable parameters

When looking at the values for the trained parameters, we find they provide some explainability. The values are visualized in figure 5, showing the parameter values during the iterations of the SPSA optimization algorithm. The parameters quickly relax to values with maximum distances between them.

We can interpret the parameters as distances within the grammatical structure. Parameters for the same grammatical type $n^r sn^l$ for "爱", 'ai' and "恨", 'hen', and $n^r s$ for "很小", 'hen xiao' and "很高", 'hen gao' respectively are $\frac{1}{2}\pi$ apart, in line with the opposing means of the terms. This mirrors the results seen in word vector embeddings, where related meanings are close to each other and opposite meanings have a large distance under the metric employed during training.

Our results here show that it possible provide explainable grammar structures through the pregroups and have explainable word embeddings similar to standard word vector embeddings. Interpreting word vectors is an important element of understanding the behaviour of neural network based NLP [48]. While the individual vector in a word space is hard to interpret, comparisons yield meaningful insights into their relationship and comparative meaning, in similar vein to what is achieved here.

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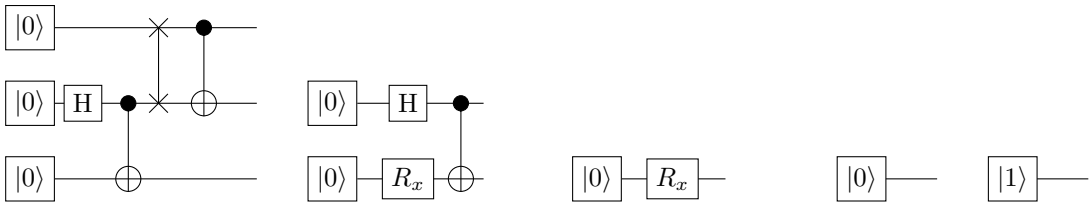


Fig. 4: The different subcircuits that are used for the different words. The trainable parameters are on the R_x rotation gates. Alice and Bob are differentiated via 0- and 1-states respectively, rather than implementing them as trainable parameters. The circuit elements are connected together to form sentences using Bell measurement on syntactically connected wires.

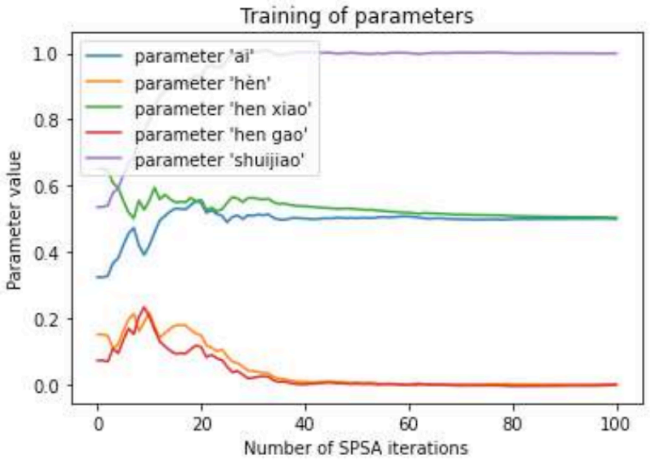


Fig. 5: Change of the five parameters expressing the verbs during the training. The parameter value is a multiple of π, such that a parameter of 1 corresponds to a rotation of π around the x axis on the respective rotation gate. The values shown here are exemplary and can be different across different runs. However, the distance between values representing opposite meanings is the same, with values of $\pi/2$ observed.

VI. Conclusion

In this work we have shown that the DisCoCat model for quantum natural language processing is applicable for the Chinese language as well, an important step towards the application of arbitrary language implementations of QNLP. We show that this can be done by adapting the pregroup system to the Chinese grammar. We achieve similar performance to English language implementations of QNLP.

Our results also indicate that the pregroup grammar approach and QNLP provide some form of interpretability towards the language model. The grammar system and its pregroup translation provide a direct connection between sentence and technical structure, and the parameters that we achieve indicate that an inference of meaning to the sentence is possible. Interpretation of large language models is an open question in the community and the presented approach may be used to expand our knowledge about language models.

VII. Future work

The experiments presented here show the general practicability and applicability of quantum computing to natural language processing. The next obvious steps are to scale up the experiments to reasonable corpus sizes. In the present experiment and those presented in literature the sentence truth label is always correct and consistent across all sentences. An exploration of the behaviour under incomplete or partially incorrect sentences would shine light on its ability to reject and disregard false labels.

This and other experiments are run on a small scale. As a consequence, only a small sample of the rich language grammar can be represented. Expanding the experiments to more complex grammar structures and larger pregroup types will shed important insight into the scaling of this approach. Relationships between words that span a large fraction of the sentence or link many different parts of a sentence together are particularly appealing areas of exploration.

We also suggest to pursue the advances in explainability further. Our results indicate that word-vector like explainability is achievable, which together with the grammar system provides a good candidate for further exploration. Quantum systems scale differently in size compared to their classical counterpart in the necessary system size, and it is worthwhile to explore how this will affect the size of word vectors. We expect a logarithmic scaling in accordance with the size of the accessible Hilbert space, but this has yet to be experimentally verified for quantum natural language processing.

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Appendix

We provide the full dataset used for training our quantum nlp system.

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Chinese sentence	Translation	Truth value
爱丽丝很小	Alice is small	false
爱丽丝很高	Alice is tall	true
爱丽丝睡觉	Alice sleeps	true
鲍勃很小	Bob is small	true
鲍勃很高	Bob is tall	false
鲍勃睡觉	Bob sleeps	false
爱丽丝爱爱丽丝	Alice loves Alice	true
爱丽丝爱鲍勃	Alice loves Bob	false
爱丽丝恨爱丽丝	Alice hates Alice	false
爱丽丝恨鲍勃	Alice hates Bob	true
鲍勃爱爱丽丝	Bob loves Alice	false
鲍勃爱鲍勃	Bob loves Bob	true
鲍勃恨爱丽丝	Bob hates Alice	true
鲍勃恨鲍勃	Bob hates Bob	false
很小的爱丽丝很小	small Alice is small	false
很小的爱丽丝很高	small Alice is tall	false
很小的爱丽丝睡觉	small Alice sleeps	false
很小的鲍勃很小	small Bob is small	true
很小的鲍勃很高	small Bob is tall	false
很小的鲍勃睡觉	small Bob sleeps	false
很高的爱丽丝很小	tall Alice is small	false
很高的爱丽丝很高	tall Alice is tall	false
很高的阿里斯睡觉	tall Alice sleeps	true
很高的鲍勃很小	tall Bob is small	false
很高的鲍勃很高	tall Bob is tall	false
很高的鲍勃睡觉	tall Bob sleeps	false
睡觉的爱丽丝很小	sleeping Alice is small	false
睡觉的爱丽丝很高	sleeping Alice is atall	false
睡觉的爱丽丝睡觉	sleeping Alice sleeps	true
睡觉的鲍勃很小	sleeping Bob is small	false
睡觉的鲍勃很高	sleeping Bob is tall	false
睡觉的鲍勃睡觉	sleeping Bob sleeps	false
爱丽丝爱很小的爱丽丝	Alice loves small Alice	false
爱丽丝爱很小的鲍勃	Alice loves small Bob	false
爱丽丝爱很高的爱丽丝	Alice loves tall Alice	true
爱丽丝爱很高的鲍勃	Alice loves tall Bob	false
爱丽丝爱睡觉的爱丽丝	Alice loves sleeping Alice	true
爱丽丝爱睡觉的鲍勃	Alice loves sleeping Bob	false
爱丽丝恨很小的爱丽丝	Alice hates small Alice	false
爱丽丝恨很小的鲍勃	Alice hates small Bob	true
爱丽丝恨很高的爱丽丝	Alice hates tall Alice	false
爱丽丝恨很高的鲍勃	Alice hates tall Bob	false
爱丽丝恨睡觉的爱丽丝	Alice hates sleeping Alice	false
爱丽丝恨睡觉的鲍勃	Alice hates sleeping Bob	false
鲍勃爱很小的爱丽丝	Bob loves small Alice	false
鲍勃爱很小的鲍勃	Bob loves small Bob	true
鲍勃爱很高的爱丽丝	Bob loves tall Alice	false
鲍勃爱很高的鲍勃	Bob loves tall Bob	false
鲍勃爱睡觉的爱丽丝	Bob loves sleeping Alice	false
鲍勃爱睡觉的鲍勃	Bob loves sleeping Bob	false

TABLE III: The full dataset of Chinese sentences used for this work. The translation is provided as an aid to the reader to cross-check the truth values.

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